

EDC BLOCK-003 (Derivation v10):

Tautology Audit and Order-of-Magnitude Sanity Check

EDC Working Document

2026-02-02 — Diagnostic before further attempts

Abstract

After NC-1 (v8) and NC-2 (v9) both returned INCONCLUSIVE due to degenerate parameter combinations, this note performs two diagnostics: **Part A (Tautology Audit):** We trace where 4D gravity concepts (G_N , $1/r$ potential, Einstein equations) first enter the EDC framework to assess whether any “derivation” of G_N is circular. **Part B (Order-of-Magnitude Check):** We set $\sigma = \sigma_0$ (a reasonable ansatz) and compute G_N^{pred} to check if EDC is “in the right universe.” **Results:** Tautology risk is MEDIUM—the 5D framework uses κ_5 without independent definition, potentially presupposing gravity. The order-of-magnitude check shows EDC *can* produce $G_N \sim 10^{-11}$ for reasonable σ , but requires fine-tuning.

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Part I

Tautology Audit

1 Purpose

A “derivation” of G_N is **tautological** if 4D gravity was already presupposed in the axioms or definitions. This audit traces the logical chain to identify potential circularity.

Tautology Criterion

A derivation is circular if:

1. G_N (or Planck mass) appears as an input parameter.
2. The 4D Einstein equations are assumed before being “derived.”
3. A $1/r$ Newtonian potential is used to define scales that later “predict” $1/r$ behavior.

2 Assumption Ledger

We trace where gravitational concepts enter the derivation chain.

Concept	Where Introduced	Tag	Tautology Risk
κ_5^2 (5D coupling)	5D EH action $S_5 \sim \kappa_5^{-2} \int R^{(5)}$	[BL]	[TAUT?] if κ_5 defined via G_5
$R^{(5)}$ (5D Ricci)	5D metric variation	[BL]	LOW (geometric)
$R^{(4)}$ (4D Ricci)	Induced on brane	[DC]	LOW (follows from $R^{(5)}$)
G_N	Matching $S_4 = \int R^{(4)}/(16\pi G_N)$	[DC]	MEDIUM (convention)
$1/r$ potential	4D Newtonian limit	[DC]	LOW if derived from $R^{(4)}$
M_{Pl}	$M_{\text{Pl}}^2 = 1/(8\pi G_N)$	[BL]	HIGH if used as input
σ (brane tension)	Brane action $S_{\text{brane}} = -\sigma \int \sqrt{-h}$	[BL]	LOW (geometric)
L (compact dim.)	Extra dimension size	[P]	LOW (geometric)

Table 1: Assumption ledger for gravitational concepts.

3 Critical Analysis

3.1 The κ_5 Question

The 5D gravitational coupling κ_5 is introduced in the bulk action:

$$S_5 = \frac{1}{2\kappa_5^2} \int d^5 X \sqrt{-g^{(5)}} R^{(5)}$$

Question: How is κ_5 defined?

- **Option 1 (non-circular):** κ_5 is a fundamental constant of the 5D theory, with dimensions $[\kappa_5^2] = L^3$, to be determined by matching to observations.

- **Option 2 (potentially circular):** κ_5 is defined via $\kappa_5^2 = 8\pi G_5$, where G_5 is the “5D Newton constant”—but what defines G_5 if not by analogy to 4D gravity?

In EDC: $\kappa_5^2 = C\sigma^{-3/4}$ is proposed, where C is unfixed. This is *not* directly circular (no G_N input), but the *interpretation* of κ_5 as “gravitational” presupposes that the 5D EH action describes gravity.

3.2 The Matching Step

The 4D Newton constant emerges from matching:

$$\frac{L}{2\kappa_5^2} = \frac{1}{16\pi G_N}$$

This is a *definition* of G_N in terms of κ_5 and L , not a derivation. The question is whether G_N so defined has the correct physical interpretation (couples to stress-energy, produces $1/r$ potential).

Assessment: This step is LOW tautology risk—it is standard dimensional reduction, not circular reasoning.

3.3 The 4D Einstein Equations

After reduction, the 4D Einstein equations emerge:

$$G_{\mu\nu}^{(4)} = 8\pi G_N T_{\mu\nu}$$

Question: Were these presupposed or derived?

Answer: Derived from the 5D equations + junction conditions. This is legitimate *if* the 5D equations are not themselves defined by requiring 4D GR in some limit.

4 Tautology Risk Assessment

Tautology Risk: MEDIUM

Findings:

- No explicit G_N^{obs} or M_{Pl} is used as input.
- The 5D EH action is assumed (not derived from more fundamental principles).
- The interpretation of κ_5 as “gravitational” is conventional, not proven.
- The matching step $G_N = \kappa_5^2/(8\pi L)$ is definitional, not circular.

Conclusion: The derivation is not *strictly* tautological, but it assumes that 5D Einstein gravity is the correct bulk theory. This is a *model assumption*, not a circularity.

Implication for BLOCK-003: The problem is not tautology but *underdetermination*—the model has one free parameter (C or equivalent) that cannot be fixed without additional input.

Part II

Order-of-Magnitude Sanity Check

5 Setup

We set $\sigma = \sigma_0$ (an ansatz) and compute G_N^{pred} to check if the order of magnitude is reasonable.

From v8:

$$G_N = \frac{C}{8\pi} \cdot \frac{\sigma^{-3/4}}{R_\xi} \quad (1)$$

We adopt the NP1 convention: $C = 8\pi$. Then:

$$G_N = \frac{\sigma^{-3/4}}{R_\xi} = \sigma^{-3/4} \cdot R_\xi^{-1} \quad (2)$$

6 Natural Units and Conversion

In natural units ($\hbar = c = 1$):

- $[G_N] = L^2/M = L^2 \cdot L = L^3$ (since $[M] = L^{-1}$)
- $[\sigma] = M/L^3 = L^{-4}$
- $[R_\xi] = L$

Check Eq. (2):

$$[G_N] = [\sigma^{-3/4}][R_\xi^{-1}] = (L^{-4})^{-3/4} \cdot L^{-1} = L^3 \cdot L^{-1} = L^2 \quad ???$$

Problem: Dimensions don't match! Let me recheck.

In SI units: $[G_N] = \text{m}^3/(\text{kg} \cdot \text{s}^2)$.

In natural units where $\hbar = c = 1$ and we measure everything in GeV:

- $[G_N] = \text{GeV}^{-2}$
- $[\sigma] = \text{GeV}^4$ (energy density)
- $[R_\xi] = \text{GeV}^{-1}$

Then:

$$[\sigma^{-3/4}/R_\xi] = \text{GeV}^{-3}/\text{GeV}^{-1} = \text{GeV}^{-2} \quad \checkmark$$

7 Numerical Estimate

7.1 Input Values

- $R_\xi = \hbar c/M_Z = 2.16 \times 10^{-18} \text{ m} \approx (91 \text{ GeV})^{-1}$
- $G_N^{\text{obs}} = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2} \approx (1.22 \times 10^{19} \text{ GeV})^{-2} = M_{\text{Pl}}^{-2}$

7.2 Required σ for Correct G_N

From Eq. (2) with $C = 8\pi$:

$$G_N = \sigma^{-3/4} R_\xi^{-1}$$

Solving for σ :

$$\sigma^{-3/4} = G_N \cdot R_\xi$$

$$\sigma = (G_N \cdot R_\xi)^{-4/3}$$

In GeV units:

$$G_N \approx (1.22 \times 10^{19})^{-2} \text{ GeV}^{-2} \quad (3)$$

$$R_\xi \approx (91)^{-1} \text{ GeV}^{-1} \quad (4)$$

$$G_N \cdot R_\xi \approx \frac{1}{(1.22 \times 10^{19})^2 \times 91} = \frac{1}{1.35 \times 10^{40}} \text{ GeV}^{-3}$$

$$\sigma = (1.35 \times 10^{40})^{4/3} \text{ GeV}^4 \approx 3.6 \times 10^{53} \text{ GeV}^4$$

7.3 Is This Reasonable?

- $\sigma^{1/4} \approx 2.4 \times 10^{13} \text{ GeV} \approx 10^{-6} M_{\text{Pl}}$
- This is a GUT-scale energy density—high but not Planckian.

Comparison: In Randall-Sundrum, $\sigma \sim k M_5^3$ where k is the AdS curvature. For $k \sim M_{\text{Pl}}$, $\sigma \sim M_{\text{Pl}}^4$, which is much larger. The EDC value $\sigma \sim 10^{53} \text{ GeV}^4$ is *smaller* than Planckian, which is a mild hierarchy.

8 Order-of-Magnitude Verdict

Order-of-Magnitude: PLAUSIBLE

Finding: To reproduce G_N^{obs} with $L = R_\xi$ and $C = 8\pi$, the required brane tension is:

$$\sigma \approx 3.6 \times 10^{53} \text{ GeV}^4 \quad \Leftrightarrow \quad \sigma^{1/4} \approx 2 \times 10^{13} \text{ GeV}$$

Assessment:

- This is GUT-scale, not Planck-scale—a mild hierarchy.
- The model is “in the right universe” but requires σ to be specified to this precision.
- No obvious contradiction with known physics.

Caveat: This is calibration, not prediction. We adjusted σ to match G_N^{obs} . True closure requires deriving σ from EDC field equations.

9 Combined Conclusions

Summary

Part A (Tautology Audit):

- Risk level: MEDIUM
- No strict circularity found
- Problem is underdetermination, not tautology

Part B (Order-of-Magnitude):

- EDC can produce correct G_N for $\sigma \sim 10^{53} \text{ GeV}^4$
- This is GUT-scale tension (plausible)
- Requires σ specification (same as before)

BLOCK-003 Status: Remains [OPEN].

The missing element is confirmed to be: **derive σ from EDC field equations** (or accept one calibration).